# Modular Refactoring and Testing of a Physics-Informed Neural Network (PINN) System for Ocean Heat Flux Prediction

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**Daily Progress Report: Trial Runs and Comparative Evaluation**

## 1 Introduction

The current phase of work focuses on conducting **trial executions and comparative evaluation of two preprocessing versions** developed as part of the modular refactoring of a Physics-Informed Neural Network (PINN) framework for ocean heat flux modeling.

Following successful restructuring of the codebase into a modular architecture, controlled trial runs were carried out to understand the **behavioral impact of alternative ocean masking and interpolation strategies**. This phase is exploratory in nature and aims to investigate *why* certain preprocessing choices influence data selection and *what* observable effects arise in spatial distributions and intermediate outputs.

At this stage, no conclusions are drawn regarding model performance or physical validity. The focus is strictly on validating pipeline behavior, understanding preprocessing effects, and building confidence in the modular execution framework.

## 2 Methodology

### 2.1 Two-Version Experimental Setup

Two versions of the preprocessing pipeline were executed under identical experimental conditions, differing only in their ocean masking and interpolation logic:

- **Version 1 (Baseline):** Closely aligned with the original preprocessing strategy.

- **Version 2 (Modified):** Implements a revised approach for ocean mask generation and NaN interpolation.

All other components, including data loading, sampling strategy, model configuration, and execution flow, were kept strictly identical to ensure a controlled and fair comparison.

## 2.2 Trial Run Configuration

The trial runs were configured as follows:

- Dataset duration: Two weeks
- Execution mode: Exploratory trial run
- Objective:
  - Validate end-to-end pipeline execution
  - Inspect spatial behavior of selected ocean points
  - Compare preprocessing outcomes across versions

These runs were not intended for training convergence or performance benchmarking.

## 2.3 Ocean Point Identification and Spatial Mapping

For both versions, the set of ocean grid points selected after preprocessing was:

- Counted to verify consistency
- Mapped onto latitude–longitude grids
- Visually inspected for geographic correctness

Spatial maps were generated to identify potential discrepancies in ocean point selection and to detect unintended exclusions or inclusions introduced by preprocessing changes.

### 2.3.1 Interpolation Behavior Visualization

To further analyze preprocessing behavior, visualization maps were generated:

- Before interpolation
- After interpolation

These visualizations were used to assess spatial continuity, smoothness, and absence of land contamination. As visual differences between the two versions were minimal, a single representative set of plots was considered sufficient for analysis.

# 3 Data, Assets, and Resources

## 3.1 Dataset

The dataset used for the trial runs consists of gridded atmospheric and ocean-related variables with the following characteristics:

- Format: NetCDF
- Temporal span: Two weeks
- Spatial structure: Regular latitude–longitude grid
- Variables include sea surface temperature and associated atmospheric parameters required for flux estimation

## 3.2 Generated Diagnostic Assets

The following assets were generated during the trial runs for qualitative inspection:

- Ocean grid point count statistics
- Spatial maps of selected ocean points
- Before-and-after interpolation maps
- Output comparison plots for both preprocessing versions

These assets were used solely for understanding pipeline behavior and not for quantitative evaluation.

## 3.3 Execution Environment

- Modular Python-based PINN framework
- Trial execution on a limited-duration dataset
- Visualization using standard scientific plotting libraries

# 4 Results and Conclusion

## 4.1 Spatial Distribution of Selected Ocean Points

Figure 1 presents the spatial distribution of ocean grid points identified after preprocessing. This visualization serves as a qualitative verification of geographic correctness and consistency across preprocessing versions.

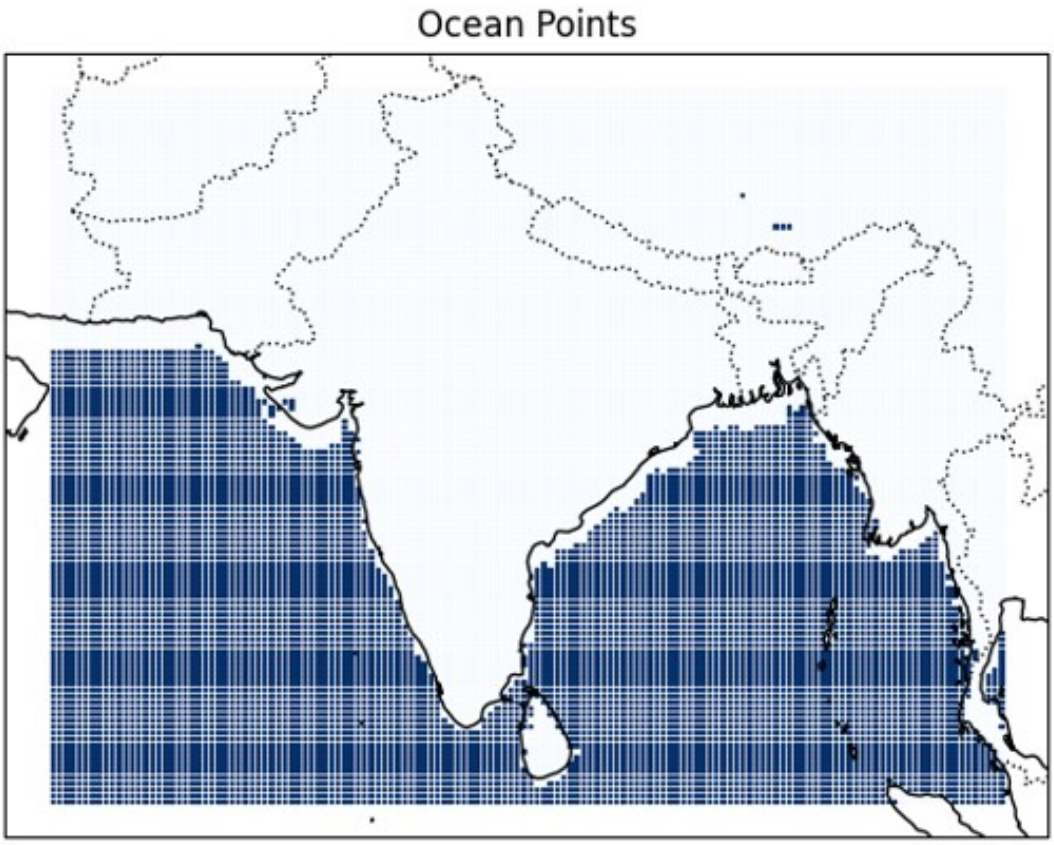


Figure 1: Spatial map of ocean grid points selected after preprocessing.

## 4.2 Interpolation Effects on Spatial Continuity

To evaluate interpolation behavior, representative spatial maps were generated before and after interpolation for selected variables.

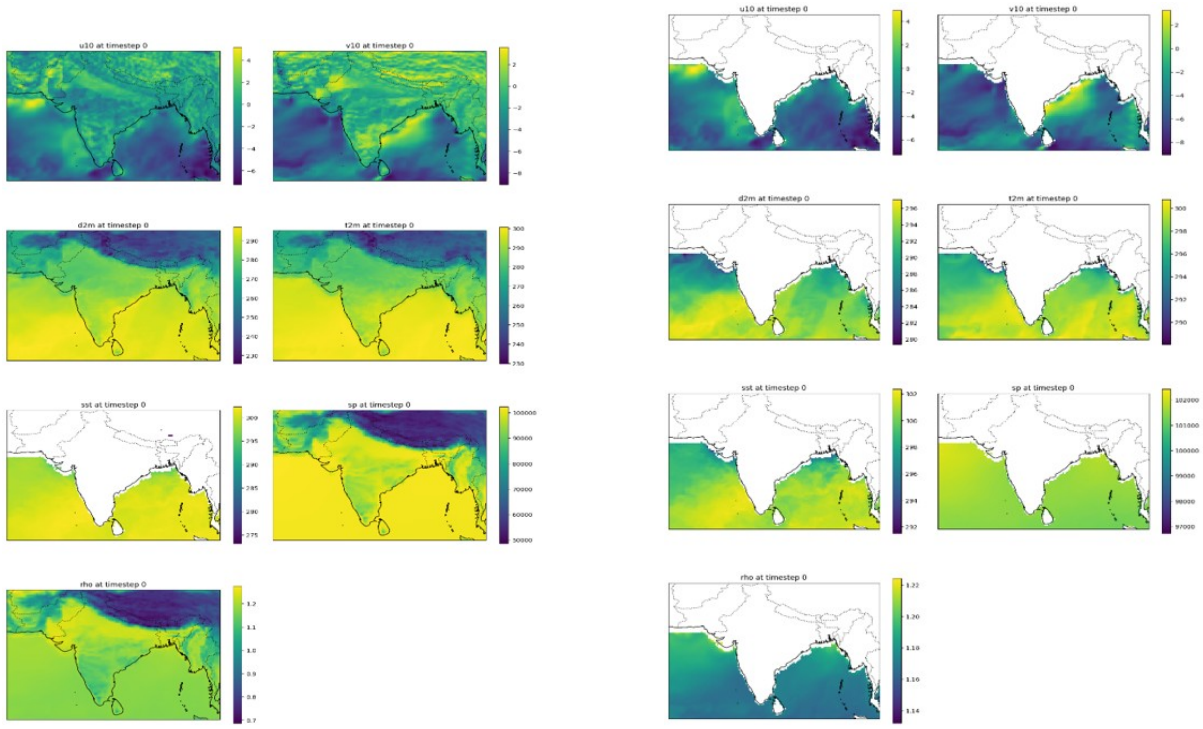


Figure 2: Representative variable maps (left) before interpolation and (right) after interpolation.

The comparison highlights smooth spatial transitions introduced by interpolation while maintaining physically plausible structures.

## 4.3 Comparative Outputs of Preprocessing Versions

Diagnostic output graphs were generated to compare preprocessing Version 1 and Version 2 under identical experimental conditions.

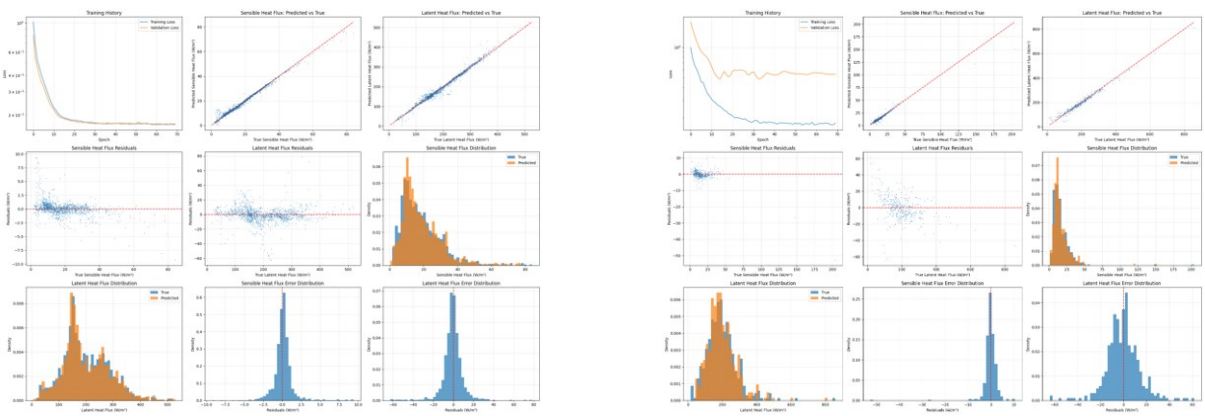

Figure 3: Comparison of diagnostic outputs: Version 1 (left) and Version 2 (right).

The outputs indicate broadly consistent behavior across both versions, with only minor localized variations observed.

## 4.4 Observations from Trial Runs

- Both preprocessing versions completed execution successfully without pipeline failure.
- Ocean grid point selection was largely consistent across both versions.
- Spatial distributions showed no major structural discrepancies.
- Interpolation behavior appeared smooth and physically reasonable in both cases.

## 4.5 Interpretation

These trial runs demonstrate that:

- The modular pipeline operates as intended.
- Small preprocessing changes can subtly influence data selection.
- Spatial visualization and point-count diagnostics are essential for understanding preprocessing effects.

This phase builds confidence in the pipeline before transitioning to structured training and physics-informed evaluation.

## 4.6 Conclusion

The exploratory trial execution phase successfully enabled controlled comparison of two preprocessing strategies within the modular PINN framework. The system is stable, observable, and ready for deeper investigation.

Subsequent work will focus on selecting a stable preprocessing strategy and progressing toward systematic training and physics-informed evaluation experiments.